EXPERIMENT-8

DATA PRE-PROCESSING & FEATURE ENGINEERING

AIM:

To implement data pre-processing and feature engineering techniques on a dataset including removing unnecessary features, feature scaling and encoding categorical variables using Python libraries

PREREQUISITES & REQUIREMENTS:
1. Computer with Python Installed
2. A dataset for Practice
3. Jupyter Notebook
4. Knowledge on Pandas & Scikit-Learn Libraries

Step 1: Import the basic libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load & Analyze the Dataset

In [ ]:
```
# Load the employee dataset
df_241UDS0096 = pd.read_excel("/content/employee dataset.xlsx")
```

In [ ]:
```
#Dataset Information
df_241UDS0096.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 223 entries, 0 to 222
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   Department          221 non-null    object 
 1   Age                 221 non-null    float64
 2   Experience_Years    221 non-null    float64
 3   Monthly_Salary      221 non-null    float64
 4   Performance_Score   221 non-null    float64
 5   Projects_Completed  223 non-null    int64  
 6   Training_Hours      223 non-null    int64  
 7   Overtime_Hours      223 non-null    int64  
 8   Satisfaction_Score  223 non-null    float64
dtypes: float64(5), int64(3), object(1)
memory usage: 15.8+ KB
```

In [ ]:
```
df_241UDS0096.head()
```

	Department	Age	Experience_Years	Monthly_Salary	Performance_Score	Projects_Completed	Training_Hours	Overtime_Hours	Satisfaction_Score
0	Finance	37.0	11.8	75682.0	87.4	10	70	21	7.0
1	R&D	21.0	4.3	81494.0	94.2	5	59	25	7.1
2	Operations	36.0	10.8	103317.0	51.3	1	46	17	7.9
3	Operations	32.0	10.0	87215.0	82.3	15	38	13	3.1
4	IT	47.0	15.4	100587.0	92.9	4	34	28	7.7

In [ ]:
```
#Check the shape
df_241UDS0096.shape
```

```
(223, 9)
```

In [ ]:
```
#Check for null values
df_241UDS0096.isnull().sum()
```

	0
Department	2
Age	2
Experience_Years	2
Monthly_Salary	2
Performance_Score	2
Projects_Completed	0
Training_Hours	0
Overtime_Hours	0
Satisfaction_Score	0

dtype: int64

In [ ]:
```
df_241UDS0096 = df_241UDS0096.dropna(how='all')

df_241UDS0096.shape
```

```
(223, 9)
```

Step 3: Remove Unnecessary Features

In [ ]:
```
# Removing unnecessary Rank column
df_241UDS0096 = df_241UDS0096.drop(
    'Overtime_Hours',
    axis=1
)

df_241UDS0096.shape
```

```
(223, 8)
```

In [ ]:
```
df_241UDS0096.columns
```

```
Index(['Department', 'Age', 'Experience_Years', 'Monthly_Salary',
       'Performance_Score', 'Projects_Completed', 'Training_Hours',
       'Satisfaction_Score'],
      dtype='object')
```

Step 4: Scale the Numerical Features

In [10]:
```
from sklearn.preprocessing import StandardScaler
```

In [ ]:
```
#Create an instance
z_scaler_241UDS0096 = StandardScaler()
```

In [ ]:
```
#Scale the numerical features
df_241UDS0096[
    ['Age', 'Experience_Years']
] = z_scaler_241UDS0096.fit_transform(
    df_241UDS0096[['Age', 'Experience_Years']]
)
```

In [ ]:
```
#After Scaling
df_241UDS0096.head(3)
```

	Department	Age	Experience_Years	Monthly_Salary	Performance_Score	Projects_Completed	Training_Hours	Satisfaction_Score
0	Finance	0.009008	-0.238963	75682.0	87.4	10	70	7.0
1	R&D	-1.667399	-1.187663	81494.0	94.2	5	59	7.1
2	Operations	-0.095768	-0.365457	103317.0	51.3	1	46	7.9

Apply Normalization using MinMaxScaler

In [14]:
```
#Import MinMaxScaler
from sklearn.preprocessing import MinMaxScaler
```

In [ ]:
```
#Create an instance
mm_scaler_241UDS0096 = MinMaxScaler()
```

In [ ]:
```
#Scale the numerical features
df_241UDS0096[
    ['Performance_Score', 'Satisfaction_Score']
] = mm_scaler_241UDS0096.fit_transform(
    df_241UDS0096[['Performance_Score', 'Satisfaction_Score']]
)
```

In [ ]:
```
#After Scaling
df_241UDS0096.head(3)
```

	Department	Age	Experience_Years	Monthly_Salary	Performance_Score	Projects_Completed	Training_Hours	Satisfaction_Score
0	Finance	0.009008	-0.238963	75682.0	0.910448	10	70	0.6250
1	R&D	-1.667399	-1.187663	81494.0	0.982942	5	59	0.6375
2	Operations	-0.095768	-0.365457	103317.0	0.525586	1	46	0.7375

Step 5: Encode the Categorical Features

Apply Integer Encoding using LabelEncoder

In [20]:
```
from sklearn.preprocessing import LabelEncoder
```

In [ ]:
```
encoder_241UDS0096 = LabelEncoder()
```

In [ ]:
```
df_241UDS0096['Department_Label'] = encoder_241UDS0096.fit_transform(
    df_241UDS0096['Department']
)
```

In [ ]:
```
df_241UDS0096.head(3)
```

	Department	Age	Experience_Years	Monthly_Salary	Performance_Score	Projects_Completed	Training_Hours	Satisfaction_Score	Department_Label
0	0	0.009008	-0.238963	75682.0	0.910448	10	70	0.6250	0
1	5	-1.667399	-1.187663	81494.0	0.982942	5	59	0.6375	5
2	4	-0.095768	-0.365457	103317.0	0.525586	1	46	0.7375	4

Apply One-Hot Encoding using pd.get_dummies()

In [ ]:
```
df_241UDS0096 = pd.get_dummies(
    df_241UDS0096,
    columns=['Department'],
    drop_first=True
)
```

In [ ]:
```
df_241UDS0096.head(3)
```

	Age	Experience_Years	Monthly_Salary	Performance_Score	Projects_Completed	Training_Hours	Satisfaction_Score	Department_Label	Department_1
0	0.009008	-0.238963	75682.0	0.910448	10	70	0.6250	0	False
1	-1.667399	-1.187663	81494.0	0.982942	5	59	0.6375	5	False
2	-0.095768	-0.365457	103317.0	0.525586	1	46	0.7375	4	False

In [ ]:
```
df_241UDS0096.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 223 entries, 0 to 222
Data columns (total 15 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Age                 221 non-null    float64
 1   Experience_Years    221 non-null    float64
 2   Monthly_Salary      221 non-null    float64
 3   Performance_Score   221 non-null    float64
 4   Projects_Completed  223 non-null    int64
 5   Training_Hours      223 non-null    int64
 6   Satisfaction_Score  223 non-null    float64
 7   Department_Label    223 non-null    int64
 8   Department_1        223 non-null    bool
 9   Department_2        223 non-null    bool
 10  Department_3        223 non-null    bool
 11  Department_4        223 non-null    bool
 12  Department_5        223 non-null    bool
 13  Department_6        223 non-null    bool
 14  Department_7        223 non-null    bool
dtypes: bool(7), float64(5), int64(3)
memory usage: 15.6 KB
```

Convert Boolean Datatypes into Integers

In [ ]:
```
bool_columns_241UDS0096 = df_241UDS0096.select_dtypes(
    include='bool'
).columns
```

In [ ]:
```
for col in bool_columns_241UDS0096:
    df_241UDS0096[col] = df_241UDS0096[col].astype(int)
```

In [ ]:
```
df_241UDS0096.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 223 entries, 0 to 222
Data columns (total 15 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   Age                 221 non-null    float64
 1   Experience_Years    221 non-null    float64
 2   Monthly_Salary      221 non-null    float64
 3   Performance_Score   221 non-null    float64
 4   Projects_Completed  223 non-null    int64  
 5   Training_Hours      223 non-null    int64  
 6   Satisfaction_Score  223 non-null    float64
 7   Department_Label    223 non-null    int64  
 8   Department_1        223 non-null    int64  
 9   Department_2        223 non-null    int64  
 10  Department_3        223 non-null    int64  
 11  Department_4        223 non-null    int64  
 12  Department_5        223 non-null    int64  
 13  Department_6        223 non-null    int64  
 14  Department_7        223 non-null    int64  
dtypes: float64(5), int64(10)
memory usage: 26.3 KB
```

In [ ]:
```
df_241UDS0096.shape
```

```
(223, 15)
```

In [ ]:
```
df_241UDS0096.to_csv(
    'processed_employee_dataset_241UDS0096.csv',
    index=False
)
```

RESULT:

The employee dataset was pre-processed by removing an unnecessary feature, scaling numerical features using StandardScaler and MinMaxScaler, and encoding the categorical Department feature using Label Encoding and One-Hot Encoding. Boolean values generated during One-Hot Encoding were converted into integers. The dataset is now transformed into a clean numerical form, ready for use in machine learning models.

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